



Traversing the activation pathway of human β_1 adrenergic receptor

Ruma Banerjee^{1*}, Souvik Karmakar², Neeraj Bharti¹, Sunitha Manjari¹, and Uddhavesh Sonavane¹

¹HPC-Medical & Bioinformatics Applications Group, Centre for Development of Advanced Computing (C-DAC),
Innovation Park, Pashan, Pune 411008

²Bioinformatics Centre, Savitribai Phule Pune University, Pune, 411007

Email: rumas@cdac.in



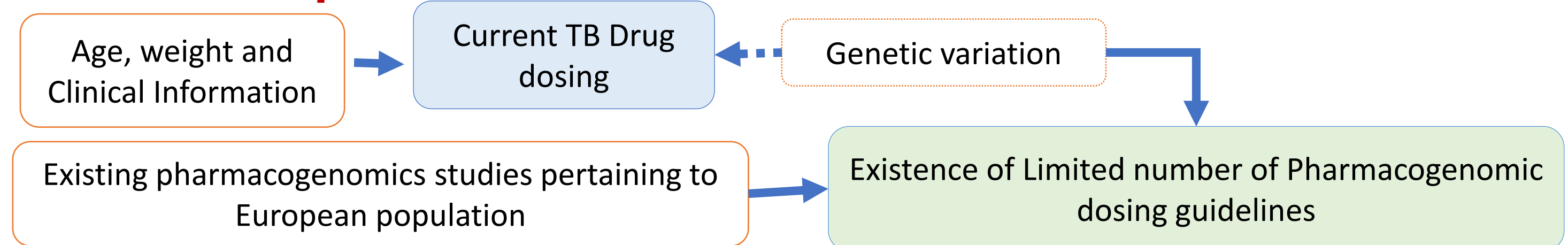
Introduction

- Tuberculosis (TB)** remains a major global health concern, with **10.8 million cases and 1.25 million deaths in 2023**, and India contributing the highest burden (~2.8 million cases)
- Standard TB treatment includes **isoniazid, rifampicin, pyrazinamide, and ethambutol**, but patient responses and adverse drug reactions vary considerably
- Pharmacogenomics** helps explain these differences by studying how genetic variations affect drug metabolism, efficacy, and toxicity
- Key genes such as **NAT2, CYP2E1, and SLCO1B1** influence the metabolism of anti-TB drugs; for example, NAT2 variants affect isoniazid clearance and hepatotoxicity risk
- Pharmacogenomic data for TB drugs are largely derived from non-Indian populations, highlighting the need for studies in genetically diverse Indian populations
- This study analyzes pharmacogenetic variation in first-line TB drug-metabolizing genes across global and Indian populations using **1000 Genomes Project** data

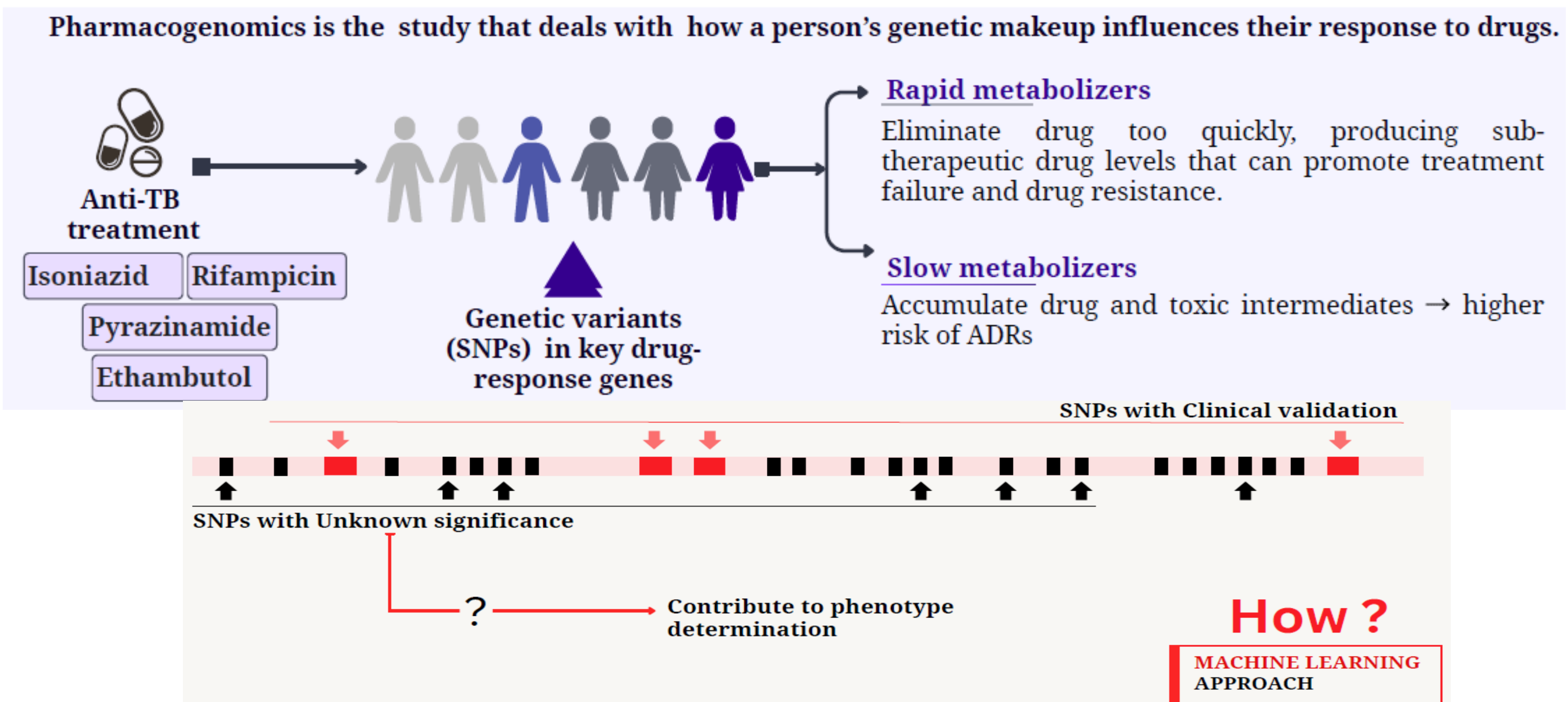
Aims and Objectives

- Examine SNPs showing significant differences in allele frequencies across populations particularly south Asian populations.
- Highlight pharmacogenomic markers associated with ADRs (adverse drug reaction), especially ATDILI (Antitubercular Drug-Induced Liver Injury).
- To generate population-specific insights that can guide the development of stratified dosing strategies in TB control
- To identify potential pharmacogenomic variants (SNPs) in genes associated with drug metabolism that influence variability in individual response to first-line anti-tuberculosis drugs

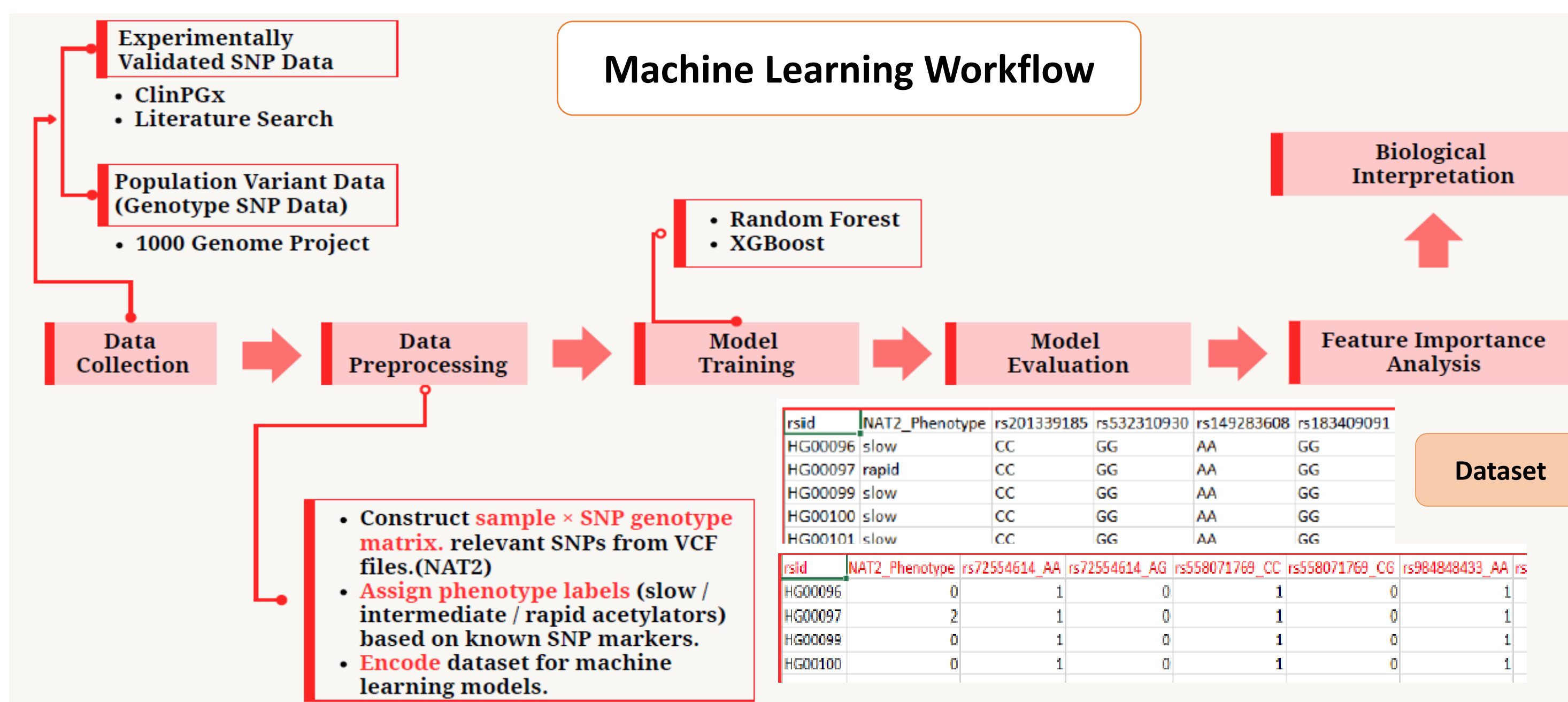
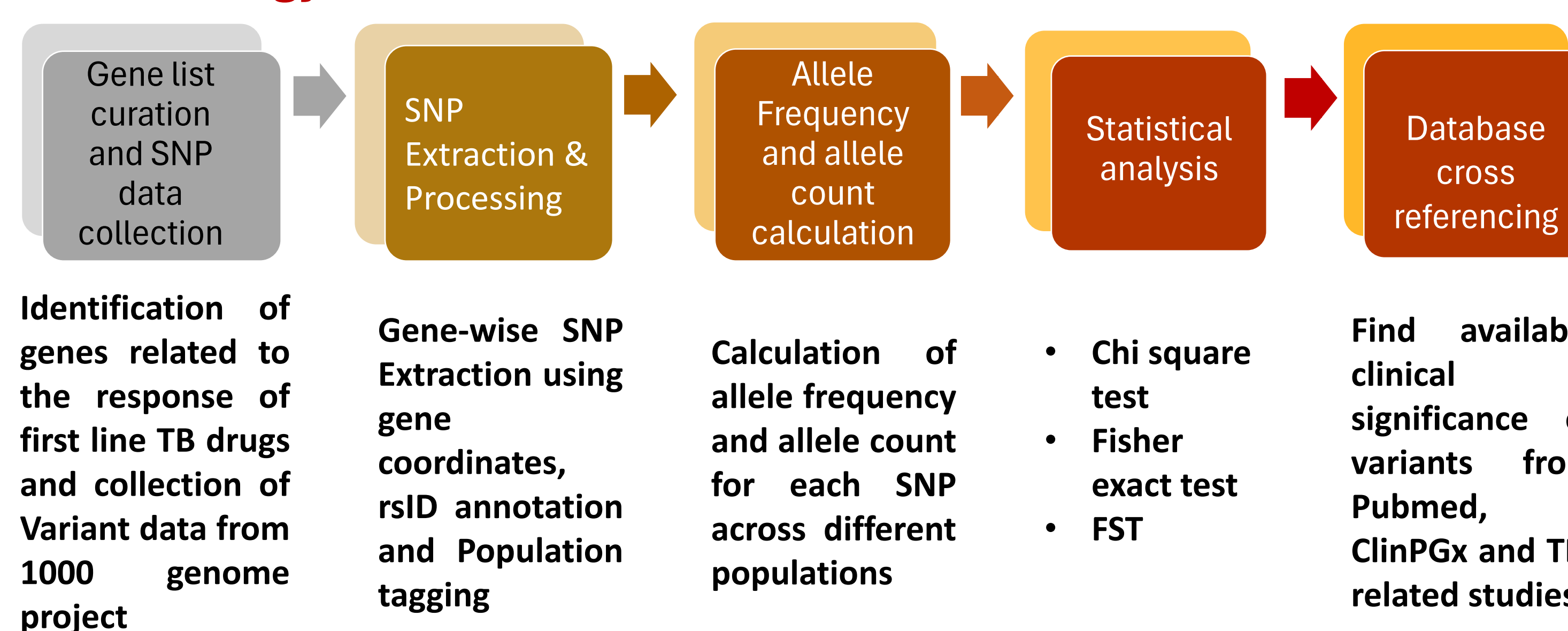
Research Gap



- South Asian populations remain severely underrepresented in pharmacogenomic research
- This study aims to bridge this gap by focusing on South Asian populations to generate evidence that can support safer, more effective, and population wise TB therapy



Methodology



rsid	NAT2_Phenotype	rs201339185	rs532310930	rs149283608	rs183409091
HG00096	slow	CC	GG	AA	GG
HG00097	rapid	CC	GG	AA	GG
HG00098	slow	CC	GG	AA	GG
HG00100	slow	CC	GG	AA	GG
HG00101	slow	CC	GG	AA	GG

rsid	NAT2_Phenotype	rs72554614_AA	rs72554614_AG	rs558071769_CC	rs558071769_CG	rs364848433_AA	rs364848433_AG
HG00096	0	1	0	1	0	1	1
HG00097	2	1	0	1	0	1	1
HG00098	0	1	0	1	0	1	1
HG00100	0	1	0	1	0	1	1

Results

- Substantial genetic diversity exists in first-line TB-drug associated genes across global and South Asian populations & also among the SAS Sub-populations
- Shared SNPs between drugs indicate overlapping metabolic and toxicity pathways, whereas unique variants (e.g., rs1968753 in CES1) highlight drug-specific mechanisms.

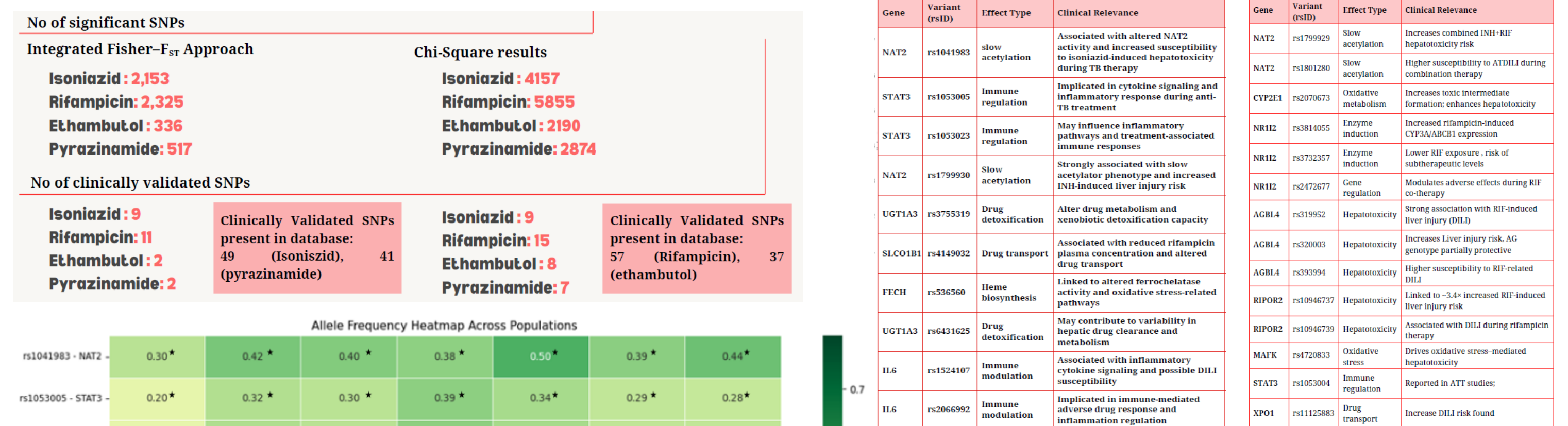
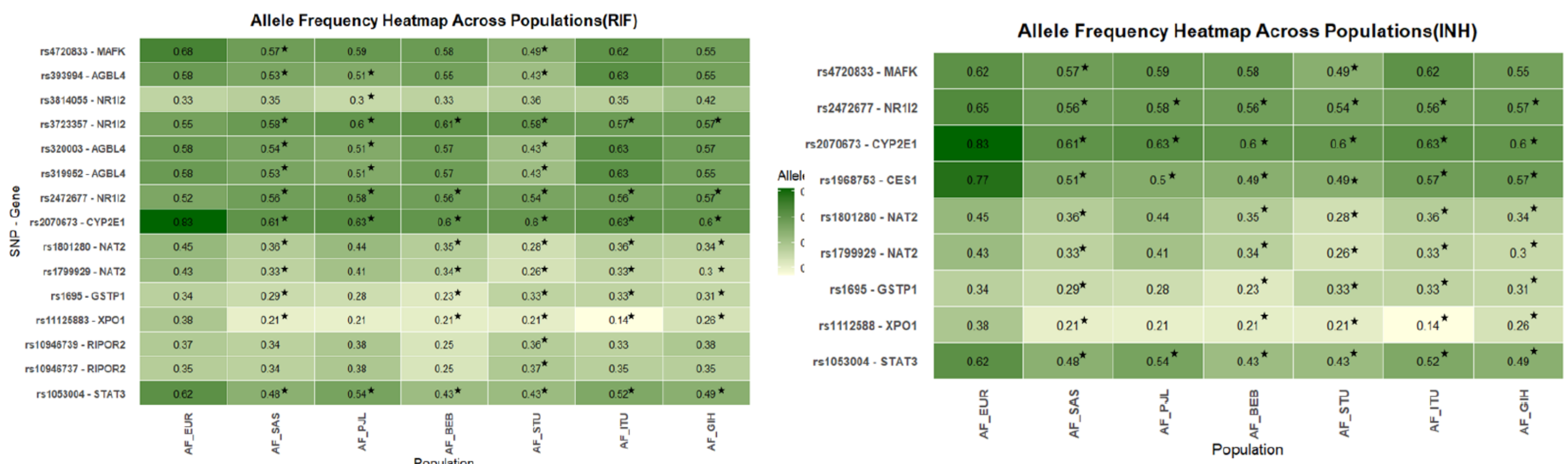


Fig.: Clinical relevance of SNPs identified in PYZ, ETB, RIF and INZ

South Asian subpopulations (PJL, GIH, STU, ITU, BEB) display distinct metabolic risk profiles, underscoring their underrepresentation in existing pharmacogenomic datasets

Fig: AF heatmap of PYZ and ETB-associated SNPs across EUR and South Asian populations



Several variants lacking prior clinical annotation were detected, indicating potential novel pharmacogenomic candidates that warrant further functional validation

ML-based identification of additional SNPs

Identification of Experimentally Validated Labelled Data

- Dataset containing experimentally validated SNPs (from pharmacogenomic studies) associated with acetylation phenotype for **NAT2** gene.
- Phenotype labels (slow, intermediate, rapid acetylators) were assigned to samples based on experimentally validated SNP markers reported in pharmacogenomic studies
- 31 SNPs and 23 diplotypes were identified from experimentally validated labelled data

Genotype matrix preparation:

- Genotype variant data obtained from the 1000 Genomes Project
- Contains SNP genotype information for multiple populations - 3202 samples & 275 SNPs (NAT2)
- Encode genotypes into numeric format (0,1)
- Phenotype Encoding: slow > 0 (1325), intermediate >1 (1215), rapid > 2 (360)
- Population-wise ML model used – Random Forest, XGBoost
- Identification of SNPs obtained from ML analysis and statistical analysis
- Model evaluation (Random Forest): Train Accuracy: 0.9991, Test Accuracy: 0.9879, CV Mean: 0.9853, Overfitting: 0.0112
- Model evaluation (XGBoost): Train Accuracy: 1.0000, Test Accuracy: 0.9966, CV Mean: 0.9961, Overfitting : 0.0034

Compared with Chi-square results					Compared with Fisher- F _{ST} approach results			
Features	Match Count (RF)	RF_Matched SNPs	Match Count (XGB)	XGB_Matched SNPs	Match Count (RF)	RF_Matched SNPs	Match Count (XGB)	XGB_Matched SNPs
1-50	16	rs45527732, rs1495744, rs45605531, rs28526210, rs1801279, rs76021804, rs1780272, rs1799929, rs1801280, rs4546703, rs28510118, rs7832071, rs4621844, rs4634684, rs13277605, rs9987109	14	rs45527732, rs1495744, rs45605531, rs28526210, rs1801279, rs76021804, rs1799929, rs1801280, rs4546703, rs28510118, rs7832071, rs45503602, rs4634684, rs9987109	19	rs1961456, rs11780272, rs1799929, rs35149886, rs9987109, rs45526838, rs45602844, rs1390358, rs1115784, rs1495744, rs923796, rs1801280, rs4546703, rs7011792, rs4621844, rs13277605, rs28510118, rs7832071, rs4634684	12	rs1495744, rs1961456, rs1799929, rs923796, rs1801280, rs4546703, rs28510118, rs1390358, rs7832071, rs1115784, rs4634684, rs9987109
51-100	3	rs45503602, rs2410556, rs45034677	2	rs4621844, rs2410556	0		1	rs4621844

Conclusions

- Comprehensive population-level pharmacogenomic analyses identified substantial genetic variability
- Interpretable AI/ML analyses successfully prioritized metabolizer-associated variants with high predictive performance
- Population-specific models revealed ancestry-dependent pharmacogenomic signatures

Acknowledgements

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- National Supercomputing Mission, MeitY, Gol

Population_model	Train Accuracy	Test Accuracy	Mean CV Accuracy	Overfitting
AFR	0.9863	0.95	0.9257	0.0332
EUR	0.994	0.99	0.9758	0.002
EAS	0.9978	1	0.9891	-0.0022
SAS	1	1	0.9978	0
AMR1	0.9953	0.98	0.9715	0.0141
AMR2	1	0.93	0.9585	0.0698

Fig.: Performance metrics of Population specific ML models

Selected References

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